Worksheet 5.4 — Reciprocal Trigonometric Functions

Grade 12 Advanced Functions (MHF4U) · Unit 5: Trigonometric Functions

The Big Ideas

$\csc\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{\cos\theta}{\sin\theta}$ each undefined where its base function is zero.

- csc $= \frac{1}{\sin}$ (undefined where $\sin = 0$); sec $= \frac{1}{\cos}$.
- cot $= \frac{\cos}{\sin}$ (undefined where $\sin = 0$).
- Evaluate the base ratio first, then reciprocate.

Core Concepts

Three more ratios

The reciprocals $\csc\theta = \dfrac{1}{\sin\theta}$, $\sec\theta = \dfrac{1}{\cos\theta}$, and $\cot\theta = \dfrac{1}{\tan\theta}$ complete the six trig functions.

Their graphs

Each reciprocal graph has vertical asymptotes wherever the original ratio is zero; cosecant and secant never take values between -1 and 1.

Where they help

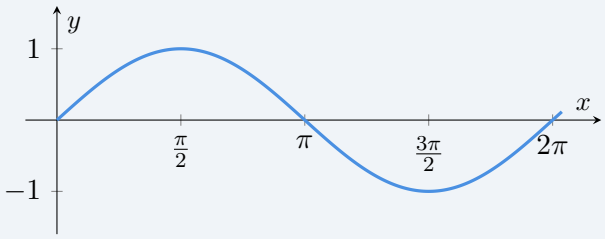
Reciprocal ratios appear in identities and in problems where a side ratio is naturally inverted.

Worked Examples

Example 1 — Cosecant

Find $\csc\frac{\pi}{4}$.

Solution. $\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2} \Rightarrow \csc = \frac{2}{\sqrt{2}} = \sqrt{2}$. The reciprocal blows up wherever sin (below) hits 0:



Example 2 — Secant

Find $\sec\frac{\pi}{6}$.

Solution. $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2} \Rightarrow \sec = \frac{2}{\sqrt{3}}$.

Example 3 — Cotangent

Find $\cot\frac{\pi}{3}$.

Solution. $\tan\frac{\pi}{3} = \sqrt{3} \Rightarrow \cot = \frac{1}{\sqrt{3}}$.

Example 4 — Cosecant

Find $\csc \frac{\pi}{6}$.

Solution. $\sin \frac{\pi}{6} = \frac{1}{2} \Rightarrow \csc = 2$.

Example 5 — Secant

Find $\sec \frac{3\pi}{4}$.

Solution. $\cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} \Rightarrow \sec = -\sqrt{2}$.

Example 6 — Cotangent

Find $\cot \frac{3\pi}{2}$.

Solution. $\frac{\cos(3\pi/2)}{\sin(3\pi/2)} = \frac{0}{-1} = 0$.

Example 7 — Secant

Find $\sec \frac{5\pi}{3}$.

Solution. $\cos \frac{5\pi}{3} = \frac{1}{2} \Rightarrow \sec = 2$.

Example 8 — Cosecant

Find $\csc \frac{3\pi}{2}$.

Solution. $\sin \frac{3\pi}{2} = -1 \Rightarrow \csc = -1$.

Example 9 — Secant of 2

Find $\sec 2\pi$.

Solution. $\cos 2\pi = 1 \Rightarrow \sec = 1$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $\csc \frac{\pi}{3}$.

Question 2

Find $\sec \frac{\pi}{4}$.

Question 3

Where does $\csc \theta$ have asymptotes?

Question 4

Find $\cot \frac{\pi}{6}$.

Question 5

Find $\sec \frac{7\pi}{6}$.

Question 6

Find $\csc \frac{5\pi}{6}$.

Question 7

Find $\sec \frac{4\pi}{3}$.

Question 8

Find $\cot \frac{2\pi}{3}$.

Question 9

Find $\csc \frac{5\pi}{3}$.

Question 10

Where does $\sec\theta$ have asymptotes?

Question 11

Find $\sec\frac{11\pi}{6}$.

Question 12

Find $\cot\pi$ (state if undefined).

Question 13 — Challenge

Find $\cot\frac{5\pi}{6}$ exactly.

Question Recap & Answer Key

Question Recap

1. Find $\csc \frac{\pi}{3}$.
2. Find $\sec \frac{\pi}{4}$.
3. Where does $\csc \theta$ have asymptotes?
4. Find $\cot \frac{\pi}{6}$.
5. Find $\sec \frac{7\pi}{6}$.
6. Find $\csc \frac{5\pi}{6}$.
7. Find $\sec \frac{4\pi}{3}$.
8. Find $\cot \frac{2\pi}{3}$.
9. Find $\csc \frac{5\pi}{3}$.
10. Where does $\sec \theta$ have asymptotes?
11. Find $\sec \frac{11\pi}{6}$.
12. Find $\cot \pi$ (state if undefined).
13. Find $\cot \frac{5\pi}{6}$ exactly.

✓ Answer Key

1. $\frac{2}{\sqrt{3}}$
2. $\sqrt{2}$
3. where $\sin \theta = 0$
4. $\sqrt{3}$
5. $-\frac{2}{\sqrt{3}}$
6. 2
7. -2
8. $-\frac{1}{\sqrt{3}}$
9. $-\frac{2}{\sqrt{3}}$
10. where $\cos \theta = 0$
11. $\frac{2}{\sqrt{3}}$
12. undefined ($\sin \pi = 0$)
13. $-\sqrt{3}$